

**Interpreting Mental Rotation Performance in Self-Described Aphantasia
through Cognitive Penetrability**

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Introduction

"The mind, it seems, is faster than even the mind's eye."
(Pylyshyn, 1981, p. 26)

Do mental images have a special status in empirical sciences, or more specifically, in cognitive science? Is the everyday expression "the mind's eye" a supporting term for assuming it to be a unique mental experience that is different from reasoning processes, thinking, or other types of mental representations? Brann characterizes a mental representation as having a mental copy of the external world, which by definition also resembles it (1991/2017). As Brann argues, the traditional approach of cognitive science is to understand this representation as being computational, which is to say that it is symbolic (i.e., an object is symbolized by something else) and formal (e.g., it is described by rules). Mental imagery then becomes a prototypical candidate for employing this definition, for it involves a resembling internal model of the outside world, having properties such as being symbolic and possibly formal. The expression "the mind's eye" implies that there is an internal (mental) world that is seen by the mind, and that this internal world resembles the external world in terms of perceptual attributes. This convention is also reflected in the mental imagery debate where the discussion revolves around the format of a mental image (Pearson & Kosslyn, 2015). As per the depictive format, parts of a mental image hold similar structures to visual perception. This original claim stems from Kosslyn et al. (1979), following the interpretation of mental scanning experiments, where it was found that scanning a mental image takes longer if targeted objects on a mental map are further apart. That is, a mental image holds similar spatial distances as its corresponding external object. Kosslyn (1994) labels this as a functional space of mental images, meaning that mental images do not occupy an actual, physical space, but rather that each part of a mental image correlates (or represents) its external object. How each part is defined is an arbitrary decision, but the mental image will still hold this resemblance. Given this unique characterization, mental images hold a special position and an exceptional interest in cognitive science.

On the other hand, mental imagery might be understood in a different manner. One of Pylyshyn's earlier queries was whether mental imagery can be explained solely by some intrinsic features of a system (e.g., the aforementioned spatial distances deriving from lawful physical relations of the world) or is it rather that mental images build on external symbolic representations (1979). The descriptive format of the mental imagery debate holds that mental images can be understood as proposition-like states, similar to (but not exactly like, see Pylyshyn, 1978) how language represents ideas through words. This is also known as the tacit knowledge argument, that is, a mental image is not held on a mental display in front of the "mind's eye", but rather it is composed of implicit knowledge about the external world that can be expressed in propositional attitudes, for example in a belief (1981). Pylyshyn argues that it is not the case that a mental image depicts a scene from which the format of the mental representation is inferred, rather that prior beliefs about the world cause mental imagery experiences, also known as the cognitive penetrability hypothesis (1979). As such, it cannot be claimed with confidence that the format of a mental image is depictive since a mental image is confounded by proposition-like representations. Thus, the mind seems to be faster than the mind's eye, according to Pylyshyn (1981). As such, its role in mental representations might not be as unique as initially thought.

Cognitive science, however, traditionally moves forward with the depictive interpretation of mental images. While Pearson and Kosslyn (2015) argued that the mental imagery debate is over, emphasizing the validity of the depictive view over the propositional,

the same year a new term was introduced into scientific inquiries that has the potential to reopen (or rather expand) this debate. In 2015, Zeman et al. coined the term *aphantasia* for those people who are unable to voluntarily form visual mental images. Recent studies on aphantasia indeed use the depictive definition of mental images, either explicitly or implicitly (e.g., Milton et al., 2021; Monzel et al., 2022; Pounder et al., 2022; Zhao et al., 2022). However, studies investigating aphantasia that employ the descriptive (or propositional) mental imagery definition are close to zero. The lack of (visual) mental images is usually identified by self-reports. The most common way to assess (visual) mental imagery is the Vividness of Visual Imagery Questionnaire (VVIQ, Marks, 1973). While the VVIQ is credited to Marks, it originates from Betts' Questionnaire Upon Mental Imagery (QMI), dating back to 1909. The date is important insofar as the QMI was developed in an era of introspective psychology, emphasizing the role of trained psychology students upon reflecting their own mental representations (Betts, 1909). While the controversy around introspection whether to accept it or not was always apparent in scientific investigations, the usage of mental imagery questionnaires remained an acknowledged method in the history of cognitive sciences (Schwitzgebel, 2019). Marks' VVIQ (1973) focuses on the vividness of mental images, presenting respondents with four scenarios to visualize and rate them on how clear and vivid each mental representation is. The general finding with respect to people with aphantasia is a consistent low average score on all four parts as opposed to people without aphantasia (e.g., Milton et al., 2021; Zeman et al., 2015).

These questionnaire items are also referred to as object imagery (Blajenkova et al., 2006), for they assess scenarios to imagine as stand-alone mental representations, focusing on what forms, sizes, or colors mental images might have. Contrary to this, spatial imagery reflects combined mental representations, assessing the spatial relationship between parts of several scenes or objects. It is emphasized by this distinction that mental imagery is not a unitary experience that can be assessed on a single dimension. For example, the Object-Spatial Imagery Questionnaire (OSIQ, Blajenkova et al., 2006) consists of 30 items addressing both imagery types, asking respondents how much they agree with each statement. Further to that, people with aphantasia have been reported to perform similarly on the OSIQ's spatial imagery items as people without aphantasia but worse on the object imagery items (e.g., Dawes et al. 2020; Keogh & Pearson, 2018). This is then interpreted as while people with aphantasia may not have object imagery, they still retain some sort of spatial imagery.

These terms, however, make use of the traditional – depictive – view of mental images. For example, the attribute 'vividness' of object imagery or the attribute 'spatial relationships' of spatial imagery derive from visual perception. According to Pylyshyn (1978) this is a confusion of the properties of an object with the properties of the mental representation. For one, similarities between the perceived object and its mental representation serve evolutionary purposes; in order to survive an analogue mental representation of the external world is required to use it properly and to act on it accordingly. However, continues Pylyshyn, the depictive view makes the faulty conclusion that following this similarity the mental representation has the same properties as the external world. Yet, a closer analysis reveals key differences between visual perception and mental imagery. As Pylyshyn argues, a mental image is already an interpreted construct, whereas a perception exists in the world independent of an observer (1978). Thus, perceptions can have multiple interpretations, whereas a mental image's interpretation only depends on one person. That is, there is a limitation on the contents of a mental image, depending on what the person knows tacitly. However, the contents of a mental image do not reveal anything about its form (Pylyshyn, 1978). As such, object and spatial imagery only hold insofar the depictive view is employed, but not according to the tacit knowledge (or cognitive penetrability) argument.

Furthermore, interpreting questionnaire items depends on participants' introspective capabilities. Schwitzgebel contends the inherent difficulty in describing expressions such as 'clarity' or 'vividness' of the VVIQ (2002). People may understand these words differently

when applying them to their own mental experiences, therefore such personal judgments of mental images cannot be compared between individuals. Consequently, there is a requirement to assess mental imagery in a more objective way. Although the list of experimental paradigms on investigating object imagery is extensive (for an overview see Pearson et al., 2013), these cognitive tasks do not explicitly deal with mental image clarity or vividness. However, spatial imagery is most commonly assessed by the mental rotation paradigm, where traditionally two objects (e.g., blocks of three-dimensional cubes assembled to form different shapes) are presented and participants are asked to indicate whether the two stimuli are identical or not (Shepard & Metzler, 1971). Crucially, the objects are positioned at different angles towards one another. The robust finding of the paradigm is that the greater the rotation angle is between the two objects (i.e., an increasing angular disparity), the greater the reaction time becomes as well, following a linear trend. Traditionally, this finding is explained by participants performing a mental rotation on one of the objects, to align it with the other (Shepard & Metzler, 1971).

However, further studies showed that there might be more than one cognitive process underlying this performance. Bilge and Taylor (2017), for example, distinguishes holistic and piecemeal strategies, whereby in the former the stimuli are processed as a whole, while in the latter parts of the stimuli are analyzed in a gradual progression. Geiser et al. (2006) further distinguishes “rotators” from “nonrotators”, depending on whether the two objects are mirrored or different ones. These findings most often involve visual manipulation of the stimuli. Khooshabeh et al. (2012), for example, created fragmented shapes (segments of the three-dimensional objects were separated). Their results indicated that a holistic strategy is more prevalent for participants with good mental imagery (better performance on whole objects), whereas a piecemeal strategy is more common for participants with poor mental imagery (better performance on fragmented objects). Most importantly, Pounder et al. (2022) found that only the lowest VVIQ-scoring participants with aphantasia performed differently to people without aphantasia, in terms of lower reaction times. As for the accuracy rates, there were no significant differences between people with and without aphantasia. All together, these findings converge to indicate that people with aphantasia might employ a more analytical, piecemeal strategy, as opposed to a holistic mental rotation.

These studies have shown that stimulus attributes can affect performance on the mental rotation task. Borst et al. (2011), however, implemented different instructions on the task, to test Pylyshyn’s theory of cognitive penetrability of mental images. It was investigated whether different instruction types could elicit different mental rotation performances. The authors stated that if beliefs affect mental imagery, this will not reveal anything about the format of the mental representation. This is in line with Pylyshyn’s reasoning, for a mental image might be confounded with other, proposition-like, mental representations (1981). On the other hand, the authors also claimed that if beliefs do not affect mental imagery, then this will indicate (one possible) format of the mental representation. This is a reasonable assumption, for it involves only one mental representation (a mental image) without the other (a belief). Their results showed that irrespective of instruction type, participants’ performance on the mental rotation task followed the usual linear trend. The authors concluded that Pylyshyn’s cognitive penetrability is not applicable on mental images.

It is yet unknown whether the same findings would apply to people with aphantasia. Borst et al. (2011) did not distinguish between good and bad mental imagery performers, based on, for example, scores on the VVIQ. Therefore, the goal of this report is to examine whether different instruction types elicit different performances on the mental rotation task among those who self-describe with aphantasia. To this end, the current study aims at replicating the first experiment of Borst et al. (2011), focusing only on the traditional mental rotation task. First, participants self-described with aphantasia randomly received an instruction, followed by the mental rotation task. Participants then filled out the OSIQ (Blajenkova et al., 2006) and the VVIQ (Marks, 1973), to clarify the – traditional – mental imagery terms (object versus spatial

imagery). Two hypotheses can be established. On the one hand, if beliefs do not affect mental imagery performance, the depictive format of a mental image might be argued for. If this is the case for aphantasia, it would be tempting to claim that aphantasia does have a depictive – and spatial – mental imagery experience. If, on the other hand, the propositionalist format is granted, beliefs are expected to modify mental images. In this case aphantasia is understood as still having some sort of a mental image, only its format cannot be defined with confidence, conforming to the introspective experience of not having any visual qualities to their mental images. From these the following predictions can be made, based on the applied instructions. Three instruction types were obtained from Borst et al. (2011). The mental rotation strategy follows the traditional instructions given to participants, that is, participants are asked to mentally rotate in their “mind’s eye” the objects until they are aligned; the leap strategy refers to a quick mental transformation of the objects, without referring to any intermediate steps of a rotation; and the analytic strategy emphasizes a piecemeal process, explicitly stating not to use any mental rotation. For the full text of instructions see the appendix of Borst et al. (2011). Following the predictions of the authors, if Pylyshyn’s theory of cognitive penetrability (1979) perseveres in this task, the traditional linear relationship between angular disparity and reaction times will not hold in case of the leap and analytic conditions but will only hold in the mental rotation condition. Additionally, the analytic condition should yield better performance than the leap or mental rotation conditions, reflected by higher efficiency (i.e., lower error rates), faster processing rates (i.e., the slopes of the best-fitting lines), and lower heights of the intercepts (i.e., the reaction time at 0°). However, if mental rotation performance is not cognitively penetrable, the traditional linear trend will appear in participants’ reaction times, along with increasing angular disparities, irrespective of instruction conditions. In this case, error rates, slopes, and the heights of the intercepts are expected to be similar. Furthermore, participants with self-described aphantasia are expected to perform lower on the spatial items of the OSIQ than the object items, and to score low on the VVIQ as well.

Method

Participants

A power calculation was performed with G*Power (Faul et al., 2007). To test an interaction effect between instruction types and angular disparities, the recommended sample size was at least 40 participants with a partial eta squared of .40, a large effect as defined by Cohen (1988), see also in Appendix A.

Participants were recruited online from two Facebook groups, created for people who self-describe with aphantasia: Aphantasia (Non-Imager / Mental Blindness) Awareness Group (n.d.), and Aphantasia Support Group (n.d.). Twenty-seven participants completed the online experiment (21 females, five males, and one non-binary; mean age 47 years and five months). Participants were randomly allocated to one of the three conditions (analytic, leap, mental rotation), nine to each group. No participants were excluded as all of them performed above chance level (above 50% accuracy) on the mental rotation task. All participants provided written consent and read the information letter that described the nature of the experimental procedure, assuring voluntary participation and the right to cancel at any time during the experiment or to withdraw their data from analysis, guaranteeing anonymity and confidential data handling, and that no risks are associated with the experiment. The study, being part of the master thesis, was approved by the Umeå University. Participants did not receive any monetary compensation.

Materials

The three-dimensional objects for the mental rotation task were acquired from Ganis and Kievit (2015), consisting of 384 shapes in total. The stimuli contained two white objects

on a black background, in the middle of the screen; the left object from the fixation cross served as the baseline shape, to which the target shape to the right of the fixation cross was placed. All objects consisted of 7-11 cubes; half of the stimuli pairs were identical shapes (matching stimuli), whereas the other half were mirrored shapes via the y-axis (non-matching stimuli). The shape on the right was either in the same orientation as the left (0°), or rotated by 50° , 100° , or 150° around the y-axis. These stimuli were similar to what Borst et al. (2011) used, see also on Figure 1.

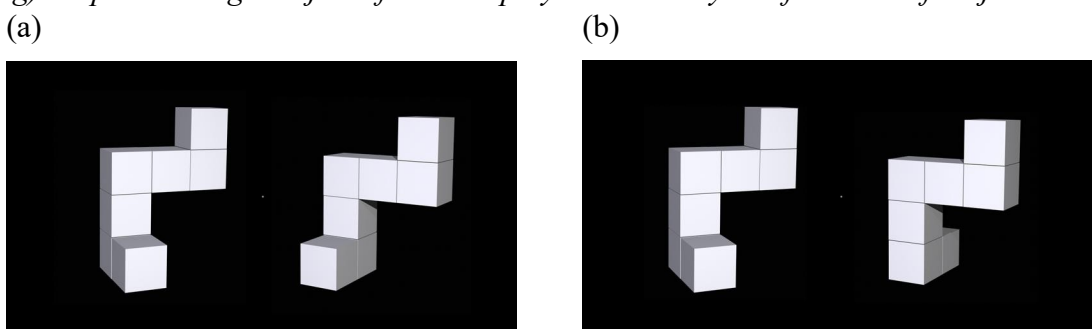
The OSIQ was acquired from Blajenkova et al. (2006). This questionnaire consists of 30 items in total: 15 measuring object imagery and 15 measuring spatial imagery, varying in a random order. Participants were asked to rate each item on a 5-point Likert scale, ranging from 1 (*totally disagree*) to 5 (*totally agree*), on how much they agree with each statement. Sample items include: “My images are very colourful and bright.” (object imagery), or “My images are more schematic than colourful and pictorial” (spatial imagery). One spatial imagery item was negatively worded (“I find it difficult to imagine how a 3-dimensional geometric figure would exactly look like when rotated.”), therefore it was reverse scored (1 = *totally agree*, 5 = *totally disagree*). A maximum possible score of 75 corresponds to either high object or spatial imagery, whereas the minimum possible score is 15 on both the object and spatial imagery items.

The VVIQ was acquired from Marks (1973). This questionnaire consists of four scenes that the participant is asked to imagine (a friend, a rising sun, a shop, and a country scene), each scene containing four items (16 items in total). Participants were asked to rate each item on a 5-point Likert scale, ranging from 1 (*No image at all, you only “know” that you are thinking of the object*) to 5 (*Perfectly clear and lively as real seeing*). While Marks originally used reversed scoring, 1 being the lowest and 5 being the highest rating, the current scoring was acquired from Zeman et al. (2015). Sample items include: “Visualize a rising sun.”, or “Think of the front of a shop which you often go to.”. No items were reverse scored. A maximum possible score of 80 is an indication of clear and vivid mental images, whereas the minimum possible score of 16 indicates no visual mental images.

The experiment with the three tasks (the mental rotation task, the OSIQ, and the VVIQ) was created in NeuroTask Scripting (NeuroTask B.V., 2021), a publicly available online platform based on JavaScript.

Figure 1

Example of a pair of three-dimensional objects (a) identical (matching) and (b) different (non-matching) shapes. The right object of each display is rotated by 50° from the left object.



Procedure

After providing the signed consent, participants received the online experiment via a URL link. Each participant first received a different instruction (analytic, leap, or mental rotation). Following this, participants were presented with two examples of the stimuli (matching and non-matching), to familiarize them with the mental rotation task. After this, 12 practice trials were followed without a time limit, and feedback was provided after each trial. Finally, the main experiment followed with a time limit of 7500 ms for each trial, and without feedback.

The process and progression of the trials of the main experiment followed the guidelines of Borst et al. (2011). Each trial commenced with a 250 ms blank screen, followed by a pair of stimuli. Participants were asked to make a judgment whether the two presented shapes are identical or not, doing so as quickly and accurately as possible. For this, participants either pressed “f” on their keyboard if the stimuli were identical, or “j” if they were not identical. Reaction time was measured from the onset of the presented stimuli until either one of the response keys was pressed. The main experiment contained two blocks of 48 trials. The four orientations (0° , 50° , 100° , 150°) were presented equally often. Half of the trials contained matching shapes. While the shapes were presented randomly, no more than three matching or non-matching trials followed consecutively, to prevent repetition effects stemming from stimulus repetition.

After the main experiment, participants completed the OSIQ and the VVIQ. At the end of the experiment, participants were asked demographic questions (age, gender, and whether they self-describe with aphantasia), whether they followed the instructions, to paraphrase the instructions how they performed the main experiment, and a debriefing question whether they inferred the purpose of the experiment. The duration of the whole experiment was less than 30 minutes, with multiple options to take a break between each part.

Results

Data analyses on reaction times only included trials with correct responses, apart from error rates that included correct and incorrect responses. While Borst et al. (2011) included only matching trials, this report also focuses on non-matching trials. Analyzing only matching trials in the mental rotation task stems from the claim that “the angle through which the “different” three-dimensional shapes must be rotated to achieve congruence is not strictly defined” (Metzler & Shepard, 1974, p. 36). That is, it could be the case that a participant starts mentally rotating a non-matching stimulus, then realizing the mismatch, she perhaps tries out another rotation, therefore the total mental rotation angle might be more than it was expected. However, Metzler and Shepard also point out that analyzing non-matching stimuli is still necessary, since participants can still make a judgment by “partial alignment” (1974, p. 53). That is, once recognized that the shapes are different, participants can infer the difference between the two shapes without any further processing time. As such, the traditional linear trend between angular disparity and reaction time might still hold for non-matching stimuli. In fact, some recent studies do include non-matching stimuli in their analyses (e.g., Geiser et al., 2006; Zhao et al., 2022), while others do not (e.g., Khooshabeh et al., 2012; Zhao & Della Sala, 2018).

Following the guidelines of Borst et al. (2011), outliers were identified as reaction times being greater than 2 *SDs* for each angle per participant. Outliers occurred on 4.6% of the matching trials, and 3.1% of the non-matching trials. Once these were removed, the mean reaction time for each angular disparity (0° , 50° , 100° , 150°) was computed for each participant, separately for the matching and non-matching objects.

Data analysis followed the same steps as it was reported in Borst et al. (2011), with two differences. First, matching and non-matching trials on the mental rotation task were analyzed

separately. Second, due to the low sample size ($N = 27$), parametric assumptions were violated in almost all cases, thus non-parametric tests were conducted, with the exception of the main analysis. Data preprocessing was performed in Microsoft Excel, descriptive and statistical analyses were executed in JASP (JASP Team, 2023), and graphs were created using GraphPad Prism (GraphPad Software, 2023).

First, each condition (analytic, leap, mental rotation) was analyzed separately whether reaction times differed significantly for the different angles, and whether reaction times correlated with angular disparity. Next, the main analysis investigated an interaction effect between instruction types and angular disparity, and whether reaction times were a linear function of angular disparity¹. Additionally, error rates, steepness of the slopes of the best-fitting lines, and the heights of the intercepts (the reaction time at 0°) were calculated, to verify the comparability of each instruction condition.

Finally, analysis on the questionnaire items (the OSIQ and the VVIQ) was performed separately, to verify significance levels. In addition, a qualitative check on the responses of participants, whether they followed the instructions and whether they followed the proper steps, showed that all participants performed according to their assigned instruction. Additionally, none of the participants inferred the purpose of the experiment, and on average participants completed the online experiment within 20 minutes, apart from two participants who had to restart the experiment due to computer issues. That is, participants on average were able to complete the experiment in one run, without major distractions.

The Mental Rotation Task

Matching Stimuli

This section covers the results of matching stimuli, while the next one covers the results of non-matching stimuli. A Friedman's repeated measures analysis of variance (ANOVA) showed that reaction times increased significantly with increasing angular disparity in all three conditions, with $\chi^2(3) = 13.652$, $p = .003$, $w = .51$, for the analytic condition; with $\chi^2(3) = 23.13$, $p < .001$, $w = .86$, for the leap condition; and with $\chi^2(3) = 21.13$, $p < .001$, $w = .78$, for the mental rotation condition. Conover's post hoc comparisons showed significant increase in reaction times with the following p values (Bonferroni corrected). For the analytic condition, with $p = .010$, between angles 0° and 100°. For the leap condition, with $p = .009$, between angles 0° and 100°, and with $p = .001$, between angles 0° and 150°. For the mental rotation condition, with $p = .006$, between angles 0° and 100°, and with $p = .004$, between angles 0° and 150°. The rest of the comparisons were non-significant, with p values ranging from .114 to 1.000. See details of JASP output in Appendix B. These findings indicate that a significant increase of reaction times with increasing angular disparity was less apparent in the analytic condition, as opposed to the other two conditions, also shown on Figure 2.

Next, correlations were analyzed for each condition. Reaction times and angular disparity were significantly correlated as follows. For the analytic condition, with $r_b = .61$, $p = .025$, between angles 0° and 100°, with $r_b = .56$, $p = .045$, between angles 0° and 150°, and with $r_b = .61$, $p = .025$, between angles 100° and 150°. For the leap condition, with $r_b = .72$, $p = .006$, between angles 0° and 100°, with $r_b = .61$, $p = .025$, between angles 0° and 150°, with $r_b = .56$, $p = .045$, between angles 50° and 150°, and with $r_b = .78$, $p = .002$, between angles 100° and 150°. For the mental rotation condition, with $r_b = .67$, $p = .013$, between angles 50° and 100°. The rest of the correlations were non-significant, with r_b values ranging from .28 to .50, p values ranging from .358 to .075. See details of JASP output in Appendix C. These results

¹ Borst et al. (2011) reported polynomial contrasts separately for each condition. However, the current report performed a non-parametric Friedman's test for these analyses, therefore Conover's post hoc comparisons were conducted (see next page). As such, only the main analysis with the mixed ANOVA contains polynomial contrasts.

indicate that linear correlations between angular disparity and reaction time were the most apparent in the leap condition, whereas the least apparent in the mental rotation condition.

For the main analysis, a mixed ANOVA was conducted with the four angular disparities as a within-subjects factor, and instruction type as a between-subjects factor. Levene's test for equality of variances was met in all angular disparities, with p values ranging from .189 to .781. Mauchly's test indicated that the assumption of sphericity had not been violated either, with $\chi^2 = 10.17$, $p = .071$. While there was a significant main effect of angular disparity on reaction times, $F(3, 72) = 49.24$, $p < .001$, $\eta_p^2 = .67$, there was no significant main effect of instruction type on reaction times, $F(2, 24) = 0.11$, $p = .894$, $\eta_p^2 = .01$. Furthermore, there was no significant interaction effect between angular disparity and instruction type, $F(6, 72) = 0.81$, $p = .565$, $\eta_p^2 = .06$. Polynomial contrasts revealed that there were significant linear and quadratic trends between angular disparity and reaction times, irrespective of instruction conditions, $F(1, 24) = 81.57$, $p < .001$, $\eta_p^2 = .77$; $F(1, 24) = 36.00$, $p < .001$, $\eta_p^2 = .60$, respectively. However, the linear trend accounted for 84.85% of the total variance. All together, these results indicate that instructions did not affect mental rotation performance, while reaction times followed both a linear and a minimal quadratic trend along increasing angular disparity, irrespective of instructions. Figure 2 shows the mean reaction times for each angular disparity, depending on instruction types. Table 1 shows further details on means and standard errors.

Figure 2

Mean reaction times for different angular disparities of matching stimuli, depending on instruction types

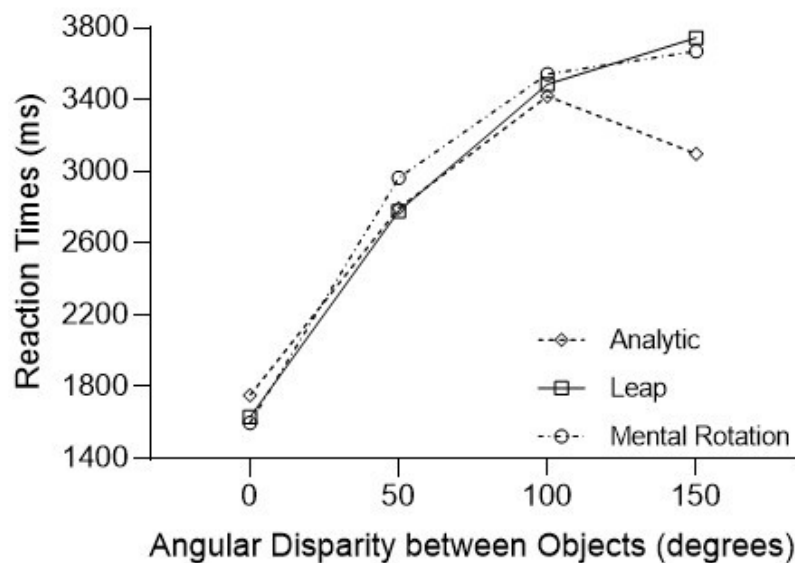


Table 1

Mean reaction times for different angular disparities of matching stimuli (ms), and standard errors of the mean (SE), for different types of instructions

Type of Instruction	Angular Disparity between Objects (Degree)			
	0	50	100	150
Analytic				
<i>M</i>	1754	2796	3420	3098
<i>SE</i>	205.01	390.58	530.46	508.62
Leap				
<i>M</i>	1631	2780	3485	3744
<i>SE</i>	147.69	310.20	339.14	307.03
Mental Rotation				
<i>M</i>	1598	2965	3542	3670
<i>SE</i>	103.42	302.27	263.93	338.71

In addition, a non-parametric equivalent of a one-way ANOVA was conducted on participants' error rates, depending on instruction types. As per the Kruskal-Wallis test, error rates did not differ significantly between the instruction types, $H(2) = 3.64$, $p = .162$, $\eta^2 = .07$. That is, error rates were comparable across the instruction conditions; the median error rates were $Mdn = 0.21$ in the analytic condition, $Mdn = 0.23$ in the leap condition, and $Mdn = 0.12$ in the mental rotation condition.

Furthermore, slopes were calculated following the guidelines of Borst et al. (2011). Slopes of the best fitting lines were calculated for each participant, which were then averaged across each instruction condition. A non-parametric equivalent of a one-way ANOVA was conducted on participants' slopes, depending on instruction types. The Kruskal-Wallis test showed that there was no significant difference in participants' slopes, $H(2) = 2.00$, $p = .369$, $\eta^2 = -.00$. The median steepness of the slopes was comparable in all instruction conditions, $Mdn = 19.40$ ms/degree in the analytic condition, $Mdn = 14.77$ ms/degree in the leap condition, and $Mdn = 12.64$ ms/degree in the mental rotation condition. That is, the average speed by which participants performed the mental rotation task was not affected by different instructions. Medians and median absolute deviations for error rates (ERs) and slopes are shown in Table 2.

Table 2

Median error rates (%), median slopes of the best-fitting lines (ms/degree) of matching stimuli, and median absolute deviations (MAD), for different types of instructions

Type of Instruction	ERs	Slopes
Analytic		
<i>Mdn</i>	21.3	19.40
<i>MAD</i>	0.07	6.78
Leap		
<i>Mdn</i>	22.9	14.77
<i>MAD</i>	0.06	4.05
Mental Rotation		
<i>Mdn</i>	12.2	12.64
<i>MAD</i>	0.07	4.30

Finally, the mean reaction times for the 0° angular disparity were used as the heights of the intercepts. As per the Kruskal-Wallis test (a non-parametric equivalent of a one-way ANOVA), instruction type did not affect significantly the height of the intercepts, $H(2) = 0.07$, $p = .967$, $\eta^2 = -.08$. The median heights of the intercepts were comparable; $Mdn = 1500$ ms for the analytic condition, $Mdn = 1656$ ms for the leap condition, and $Mdn = 1645$ ms for the mental rotation condition.

Non-Matching Stimuli

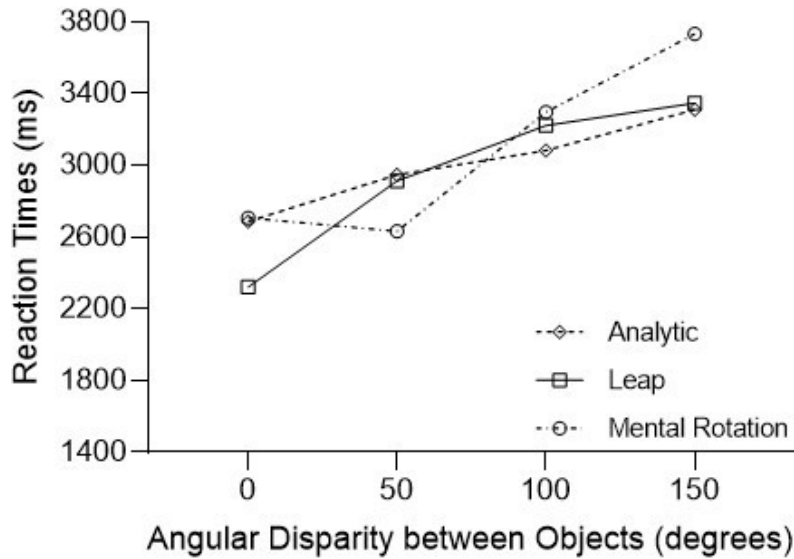
A Friedman's repeated measures ANOVA showed that reaction times increased significantly with increasing angular disparity only in two conditions, with $\chi^2(3) = 12.60$, $p = .006$, $w = .47$, for the leap condition; and with $\chi^2(3) = 22.20$, $p < .001$, $w = .82$, for the mental rotation condition. However, no significance level was reached in the analytic condition, with $\chi^2(3) = 6.07$, $p = .108$, $w = .23$. Conover's post hoc comparisons showed significant increase in reaction times with the following p values (Bonferroni corrected). For the leap condition, with $p = .021$, between angles 0° and 100°. For the mental rotation condition, with $p = .021$, between angles 0° and 150°, with $p = .021$, between angles 50° and 100°, and with $p = .006$, between angles 50° and 150°. The rest of the comparisons were non-significant, with p values ranging from .076 to 1.000. See details of JASP output in Appendix D. These findings indicate that the analytic condition yielded the “flattest” reaction times with increasing angular disparity, while the mental rotation condition showed the most apparent increase in reaction time for non-matching stimuli, also shown on Figure 3.

Next, correlations were analyzed for each condition. Reaction times and angular disparity were significantly correlated as follows. For the analytic condition, with $\tau_b = .72$, $p = .006$, between angles 50° and 100°, and with $\tau_b = .61$, $p = .015$, between angles 100° and 150°. For the leap condition, with $\tau_b = .67$, $p = .013$, between angles 0° and 50°, with $\tau_b = .67$, $p = .013$, between angles 50° and 100°, and with $\tau_b = .56$, $p = .045$, between angles 100° and 150°. As for the mental rotation condition, correlations between all angles were significant, with τ_b values ranging from .83 to .61, p values ranging from $< .001$ to .025. The rest of the correlations in the analytic and leap conditions were non-significant, with τ_b values ranging from -.06 to .44, p values ranging from .919 to .119. See details of JASP output in Appendix E. These results indicate that linear correlations between angular disparity and reaction time for non-matching stimuli were the most apparent in the mental rotation condition, whereas the least apparent in the analytic condition.

For the main analysis of non-matching stimuli, a mixed ANOVA was conducted with the four angular disparities as a within-subjects factor, and instruction type as a between-subjects factor. Levene's test for equality of variances was met in all angular disparities, with p values ranging from .419 to .921. Mauchly's test indicated that the assumption of sphericity had not been violated either, with $\chi^2 = 10.33$, $p = .067$. While there was a significant main effect of angular disparity on reaction times, $F(3, 72) = 19.59$, $p < .001$, $\eta_p^2 = .45$, there was no significant main effect of instruction type on reaction times, $F(2, 24) = 0.08$, $p = .920$, $\eta_p^2 = .01$. Furthermore, there was no significant interaction effect between angular disparity and instruction type, $F(6, 72) = 1.72$, $p = .128$, $\eta_p^2 = .13$. Polynomial contrasts revealed that there was a significant linear trend between angular disparity and reaction times, irrespective of instruction conditions, $F(1, 24) = 35.67$, $p < .001$, $\eta_p^2 = .60$. The quadratic trend was not significant, $F(1, 24) = .001$, $p = .982$, $\eta_p^2 = .00$. All together, these results indicate that instructions did not affect mental rotation performance of non-matching stimuli either (with the possible exception of the analytic condition), while reaction times followed a linear trend along increasing angular disparity, irrespective of instructions. Figure 3 shows the mean reaction times for each angular disparity, depending on instruction types. Table 3 shows further details on means and standard errors.

Figure 3

Mean reaction times for different angular disparities of non-matching stimuli, depending on instruction types

**Table 3**

Mean reaction times for different angular disparities of non-matching stimuli (ms), and standard errors of the mean (SE), for different types of instructions

Type of Instruction	Angular Disparity between Objects (Degree)			
	0	50	100	150
Analytic				
M	2688	2948	3084	3312
SE	246.57	289.79	310.24	200.64
Leap				
M	2321	2914	3224	3349
SE	292.92	253.27	331.54	333.50
Mental Rotation				
M	2708	2634	3298	3735
SE	213.11	287.41	285.75	281.34

In addition, a non-parametric equivalent of a one-way ANOVA was conducted on participants' error rates, depending on instruction types. As per the Kruskal-Wallis test, error rates did not differ significantly between the instruction types, $H(2) = 3.64$, $p = .162$, $\eta^2 = .07$. That is, the median error rates were comparable across the instruction conditions, with $Mdn = 0.14$ in the analytic condition, $Mdn = 0.22$ in the leap condition, and $Mdn = 0.06$ in the mental rotation condition. This is consistent with the results on the error rates of matching stimuli.

Furthermore, slopes were calculated similarly as previously reported for the matching stimuli. A non-parametric equivalent of a one-way ANOVA was conducted on participants' slopes, depending on instruction types. The Kruskal-Wallis test showed that there was no significant difference in participants' slopes, $H(2) = 0.52$, $p = .772$, $\eta^2 = -.06$, corresponding to the findings of matching stimuli. The median steepness of the slopes was comparable in all instruction conditions, $Mdn = 6.90$ ms/degree in the analytic condition, $Mdn = 8.00$ ms/degree in the leap condition, and $Mdn = 6.24$ ms/degree in the mental rotation condition. That is, the

average speed by which participants performed the mental rotation task was not affected by different instructions for the non-matching stimuli either. Medians and median absolute deviations for error rates (ERs) and slopes are shown in Table 4.

Table 4

Median error rates (%), median slopes of the best-fitting lines (ms/degree) of non-matching stimuli, and median absolute deviations (MAD), for different types of instructions

Type of Instruction	ERs	Slopes
Analytic		
<i>Mdn</i>	14.3	6.90
<i>MAD</i>	0.12	4.93
Leap		
<i>Mdn</i>	21.7	8.00
<i>MAD</i>	0.09	3.64
Mental Rotation		
<i>Mdn</i>	6.40	6.24
<i>MAD</i>	0.04	1.07

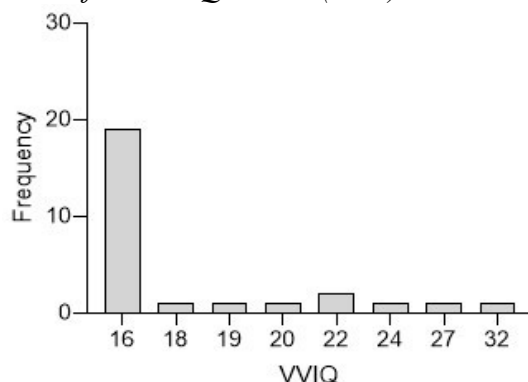
Finally, the mean reaction times for the 0° angular disparity were used as the heights of the intercepts, similarly as with matching stimuli. A non-parametric equivalent of a one-way ANOVA was conducted on the height of the intercepts, depending on instruction types. As per the Kruskal-Wallis test, instruction type did not affect significantly the height of the intercepts, $H(2) = 2.13$, $p = .345$, $\eta^2 = .01$. Similarly to what was found for matching stimuli, medians were comparable; *Mdn* = 2664 ms for the analytic condition, *Mdn* = 1913 ms for the leap condition, and *Mdn* = 2768 ms for the mental rotation condition.

The OSIQ and the VVIQ

A non-parametric equivalent of a paired samples t-test (Wilcoxon's signed-rank test) on the object and spatial components of the OSIQ showed that the spatial components (*Mdn* = 37) were significantly higher than the object components (*Mdn* = 19), $W = 26.50$, $p < .001$, $r_B = .86$. As it was predicted, people with aphantasia scored higher on spatial imagery items than on object imagery items. An additional Wilcoxon signed-rank test on the VVIQ showed that the median score (*Mdn* = 16) was also significant, $Z = 93.92$, $p < .001$, $d = 18.07$. Figure 4 shows the frequencies of total scores, 16 (the lowest possible) as being the most common case.

Figure 4

Distribution of the VVIQ scores (total)



Discussion

The goal of this study was to investigate the possibility of the effects of instruction-induced beliefs on the mental rotation task in people with self-described aphantasia. Considering only matching shapes, individual analyses of each condition showed some inconsistencies; while the analytic condition had the least reaction time increase, the mental rotation condition showed the least correlations between reaction time and angular disparity. These findings do not replicate Borst et al. (2011) and would suggest (prematurely) that instruction-induced beliefs affected mental rotation performance. More importantly, however, a non-significant interaction effect between angular disparity and instruction type on participants' reaction times showed that different instruction conditions did not influence mental rotation performance. Furthermore, reaction times on average also followed the traditional linear trend, irrespective of instructions. Additionally, efficiency, speed of processing, and the heights of the intercepts were comparable between instruction types. These results replicate what Borst et al. (2011) found, and are partially in line with the current predictions, that is, performance in the analytic condition did not differ substantially from the other conditions.

Next, following the guidelines of Metzler and Shepard (1974), the main result showed that instruction types did not interact significantly with angular disparities for non-matching shapes either. Although the individual analyses of conditions showed that reaction times for the analytic instruction did not increase significantly, and that it yielded the least number of significant correlations with angular disparity, the main analysis contrasts this finding. In fact, a traditional linear trend on average was also found for these shapes, and efficiency, the speed of processing, and the heights of the intercepts were comparable, irrespective of instructions. These results – taken all together – are in line with the previous predictions. Interestingly, the results of Metzler and Shepard (1974) showed that the largest angular disparity (180°) was processed slower than the smaller angular disparities, not conforming to the expected linear trend. While the current study did not replicate this finding, it could be argued – with caution – that the analytic condition yielded a slightly different processing time than the other two conditions (especially the mental rotation condition).

Taken all together, the evidence suggests that instruction-induced beliefs did not have an impact on mental rotation performance for those who self-describe with aphantasia. The traditional linear trend between reaction times and angular disparity supports the claim that participants mentally rotated (i.e., *possibly* visualized each intermediate step between different angles) the shapes, which would indicate that mental images have intrinsic features, irrespective of tacit knowledge (Borst et al., 2011). This claim would then be against Pylyshyn's thesis of cognitive penetrability (1979), according to which beliefs about the external world would modify mental imagery experiences, which also manifests in the propositional definition of mental images. Contrary to this, the depictive view contends that mental images hold similar structures to their corresponding external objects (Kosslyn et al., 1979). If this correspondence stands, then the claim that mental images are unique mental representations that hold a special place in cognitive science could be warranted. Following Brann's definition of a mental representation (1991/2017), a mental image then becomes symbolic, for it symbolizes the external world, but also – in a way – formal, meaning that it follows spatial rules deriving from the external world.

The overall findings are in line with previous research on aphantasia. As for the mental imagery questionnaires, the low scores on the VVIQ are in accord with Zeman et al. (2015), who coined the term 'aphantasia' for those who cannot form visual mental images. In fact, the authors found the same median total score (16) for their participants with aphantasia as this study reported (see also on Figure 4). Furthermore, participants scored higher on spatial imagery than on object imagery items, which is consistent with the results of Keogh and Pearson (2018), and Dawes et al. (2020). While the participants of the current study were

volunteers who self-described with aphantasia, such questionnaires can verify the possibility of them not having any visual qualities to their mental images. Thus, the mental rotation paradigm becomes a key element to investigate their mental representations in an objective way. While Bilge and Taylor (2017) distinguished two mental rotation strategies (holistic and piecemeal), and Gesier et al. (2006) further distinguished “rotators” from “nonrotators”, the key finding that is in accord with the results of this study was established by Pounder et al. (2022). The authors found that people with aphantasia performed similarly on the mental rotation task than people without aphantasia. Caution must be taken, however, as the current study did not include control participants who would score high on, for example, the VVIQ. Therefore, claims between aphantasia and its absence (i.e., an intact mental imagery) cannot be made. Moreover, while Khooshabeh et al. (2012) found that bad imagers performed a piecemeal strategy on the mental rotation task, contrary to good imagers with a more holistic strategy, a claim was made earlier that people with aphantasia might employ more of an analytical process during the mental rotation task, as opposed to a traditional – visual mental imagery-based – process. Only little support was found to this claim, in terms of the fact that non-matching shapes were not processed differently with increasing angular disparity in the analytic condition.

In terms of the previously established hypotheses, several interpretations emerge from these findings. As the support is stronger for claiming the validity of the depictive view of mental images, as opposed to the propositional view, it might have been the case that people with aphantasia still have certain – spatial – mental imagery. In fact, Blazhenkova and Pechenkova (2019) argued for distinguishing object aphantasia from spatial aphantasia. This would then be an indication that mental imagery cannot be understood as a unitary experience, and as such an explanation of what a mental image is would need to be more thorough. Pylyshyn was among the first ones to recognize the need for this. While it might not be the case that mental images are cognitively penetrable, as it was claimed by him in 1979, the emphasis is more on how new scientific terms are created. Certainly, the label ‘aphantasia’ is useful insofar it generates public interest and scientific research, it might have been a precocious attempt to define the diversity of such mental representations under one umbrella term. However, it could be argued against this that the investigations on the form of mental images have a long tradition both in philosophy and in psychology (for an overview see Brann 1991/2017), and most certainly that the lack of mental images had already been reported as early as 1880 by Galton. Therefore, the question will be left open whether the term ‘aphantasia’ is a fruitful addendum to the investigations of cognitive sciences.

Some limitations of the current study need to be addressed. Although the experiment aimed for excluding repetition effects in that no more than three consecutive matching (or non-matching) trials were presented, accuracy in progressive mental rotations could have been ameliorated, especially for non-matching stimuli in the analytic condition. This suspicion is reinforced by the findings of Wan et al. (2011), who showed that a trial of mental rotation of an alphabetic character was improved when the preceding trial required the same direction to orientate the stimulus. This was irrespective of stimulus repetition. Furthermore, Jost and Jansen (2022) found that actual – physical – rotation movements can improve mental rotation performance of three-dimensional objects, similar to the stimuli that were employed in the current study. The authors further contend that this training effect occurs due to an assumed visual rotation, be it either physical or mental. Therefore, it could have been the case that while with increasing angular disparity participants of this study performed slower, nonetheless practice made their performance more efficient in time.

A further limitation can be considered in terms of the nature of the sample, since participant recruitment was not random, but was based on volunteering. As it turned out,

women were overrepresented in the sample (21 out of 27)². It could be argued that mental rotation performance differs between sexes. Traditional studies on mental rotation generally find such sex differences, in terms of men outperforming women (for a meta-analysis see Voyer et al., 1995). However, Scheer et al. (2018) found no evidence for such sex differences on the traditional mental rotation task of three-dimensional objects, which also employed time limits on the task, similarly to the current study. Men and women performed equivalently, both in terms of accuracy and reaction time. For a better assessment of sex differences, future studies of mental rotation targeting people with aphantasia should recruit both genders equally.

A further suggestion for future studies would be to compare people with and without aphantasia on the mental rotation task, with the implementation of different instruction conditions. Studies on aphantasia generally do this comparison, to record cognitive differences between those who claim to have intact visual mental imagery and those who claim the opposite (e.g., Milton et al., 2021, Pounder et al., 2022). Further to that, additional cognitive processes could be investigated next to the mental rotation task. Pylyshyn himself admits that the mental rotation task might not be a good “candidate for a straightforward tacit knowledge explanation” (2002, p. 165), for it might involve other (e.g., working memory) processes irrespective of the form of a mental image. Studies such as Jacobs et al. (2018) or Monzel et al. (2022), for example, investigated visual working memory and found that people with aphantasia performed worse on such cognitive tasks than people without aphantasia. Further to that, Monzel et al. (2022) found that participants with aphantasia performed worse on verbal memory tasks (both short- and long-term), compared to people without aphantasia. Thus, expanding the scope of what other types of cognitive processes (or mental representations) might influence the mental rotation performance in people with aphantasia could become a major path for future investigations.

As debated earlier, whether the term ‘aphantasia’ is useful or not has a further practical implication. If the phenomenon is understood poorly – which is argued for by this report – it could lead to extreme claims such as that aphantasia might be a mental disorder. This is reflected in several recent papers (e.g., Fox-Muraton, 2021; Whiteley, 2021) whose authors explicitly use the term ‘disorder’. Given that people with aphantasia might have impairments of working memory (see Jacobs et al., 2018; Monzel et al., 2022), such cognitive deteriorations could indicate the possibility of having a worsened functioning in daily activities. However, the assessment of Monzel et al. (2023) showed that everyday distress in the forms of anxiety or depression can be considered too low to classify aphantasia as a mental disorder; only a small subset of people with no visual mental imagery would be required clinical treatment. As such, it remains an important path for future research to better understand what the lack of visual mental imagery might mean.

This report enumerated several of Pylyshyn’s arguments in terms of why the expression “the mind’s eye” could be misleading. While the most important of his ideas – the cognitive penetrability hypothesis (1979) – was not supported by the current experimental findings, an important note must be taken. Verbalizing the subjective experience of a mental image is inherently difficult. This is also reflected in the fact that traditional psychological introspection is a practice that was not celebrated by all scientific communities throughout history (Schwitzgebel, 2019). As it was also pointed out by Schwitzgebel earlier (2002), talking about clarity or vividness of mental images might be interpreted differently between people. Pylyshyn adds to this argument when he asserts that intuitions about mental representations might not reveal anything about its form, following the analogy with visual perception where empirical findings support the claim that even though we report on seeing a rich visual scene, “the information available to cognition from a single glance is extremely impoverished” (2002, p.

² An incidental observation was that while 55 people showed interest in the study through the Facebook groups, only eight of them were men. Why women were overrepresented in online volunteering will be left an open question for now.

159). Whether a “single glance” of a mental image provides sufficient information to cognition about its form or not will be left an open question for now. While the subjective experience of having mental images (or the lack of it) cannot be impugned, how a mental image “looks like” for its beholder remains a matter of further debate. Such arguments thus invite new avenues for future research. Given that the current report reinforces the idea that mental images hold a special status in cognitive science, it might be the case that the mind is not faster than the “mind’s eye”, after all.

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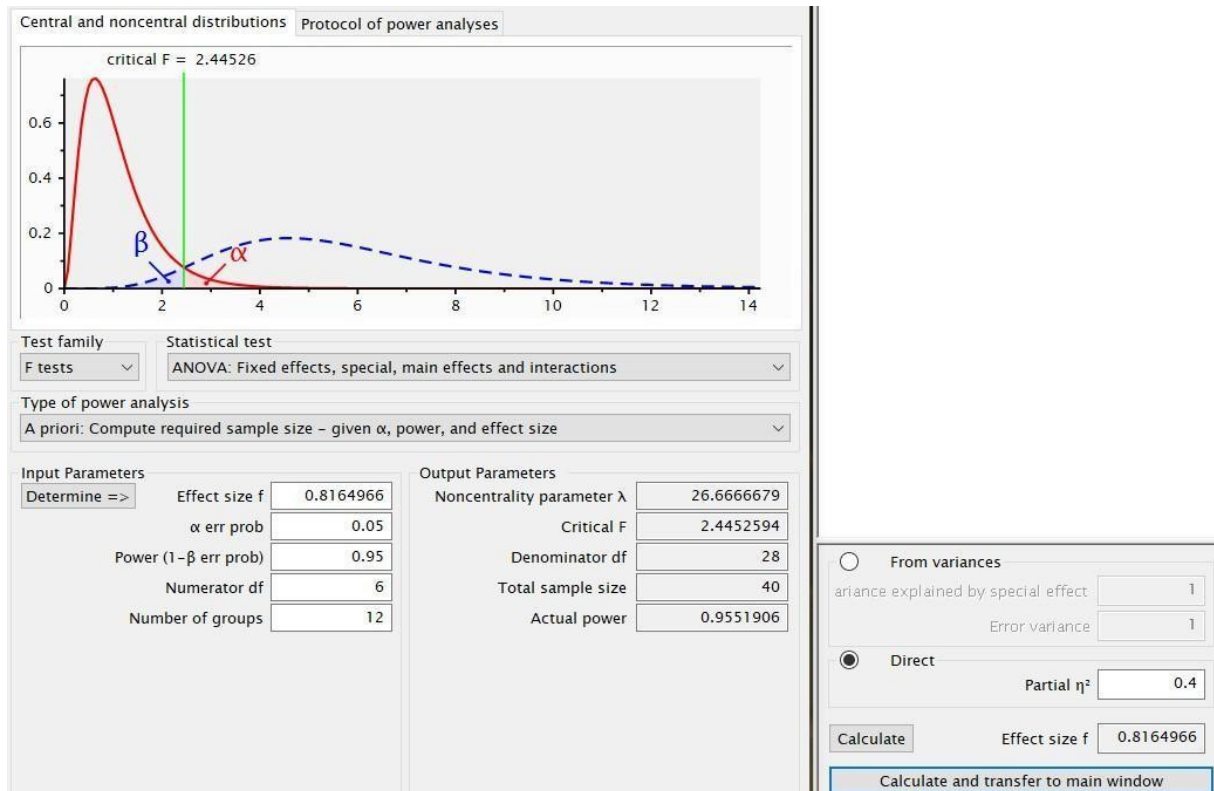
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Appendix A

G*Power Calculation



Appendix B

Conover's Post Hoc Comparisons, Matching Stimuli, Analytic Condition

Conover's Post Hoc Comparisons - Angle

		T-Stat	df	W _i	W _j	p	P _{bonf}	P _{holm}
0	50	2.352	24	11.000	24.000	0.027	0.163	0.112
	100	3.527	24	11.000	30.500	0.002	0.010	0.010
	150	2.442	24	11.000	24.500	0.022	0.134	0.112
50	100	1.176	24	24.000	30.500	0.251	1.000	0.754
	150	0.090	24	24.000	24.500	0.929	1.000	0.929
100	150	1.085	24	30.500	24.500	0.289	1.000	0.754

Note. Grouped by subject.

Conover's Post Hoc Comparisons, Matching Stimuli, Leap Condition

Conover's Post Hoc Comparisons - Angle

		T-Stat	df	W _i	W _j	p	P _{bonf}	P _{holm}
0	50	1.798	24	9.000	19.000	0.085	0.509	0.254
	100	3.596	24	9.000	29.000	0.001	0.009	0.007
	150	4.315	24	9.000	33.000	< .001	0.001	0.001
50	100	1.798	24	19.000	29.000	0.085	0.509	0.254
	150	2.517	24	19.000	33.000	0.019	0.114	0.076
100	150	0.719	24	29.000	33.000	0.479	1.000	0.479

Note. Grouped by subject.

Conover's Post Hoc Comparisons, Matching Stimuli, Mental Rotation Condition

Conover's Post Hoc Comparisons - Angle

		T-Stat	df	W _i	W _j	p	P _{bonf}	P _{holm}
0	50	1.978	24	9.000	20.000	0.060	0.357	0.238
	100	3.776	24	9.000	30.000	< .001	0.006	0.005
	150	3.955	24	9.000	31.000	< .001	0.004	0.004
50	100	1.798	24	20.000	30.000	0.085	0.509	0.238
	150	1.978	24	20.000	31.000	0.060	0.357	0.238
100	150	0.180	24	30.000	31.000	0.859	1.000	0.859

Note. Grouped by subject.

Appendix C

Correlations, Matching Stimuli, Analytic Condition

Kendall's Tau Correlations

Variable		RT_0	RT_50	RT_100	RT_150
1. RT_0	Kendall's Tau B	—			
	p-value	—			
2. RT_50	Kendall's Tau B	0.278	—		
	p-value	0.358	—		
3. RT_100	Kendall's Tau B	0.611	0.444	—	
	p-value	0.025	0.119	—	
4. RT_150	Kendall's Tau B	0.556	0.389	0.611	—
	p-value	0.045	0.180	0.025	—

Correlations, Matching Stimuli, Leap Condition

Kendall's Tau Correlations

Variable		RT_0	RT_50	RT_100	RT_150
1. RT_0	Kendall's Tau B	—			
	p-value	—			
2. RT_50	Kendall's Tau B	0.500	—		
	p-value	0.075	—		
3. RT_100	Kendall's Tau B	0.722	0.444	—	
	p-value	0.006	0.119	—	
4. RT_150	Kendall's Tau B	0.611	0.556	0.778	—
	p-value	0.025	0.045	0.002	—

Correlations, Matching Stimuli, Mental Rotation Condition

Kendall's Tau Correlations ▼

Variable		RT_0	RT_50	RT_100	RT_150
1. RT_0	Kendall's Tau B	—			
	p-value	—			
2. RT_50	Kendall's Tau B	0.444	—		
	p-value	0.119	—		
3. RT_100	Kendall's Tau B	0.333	0.667	—	
	p-value	0.260	0.013	—	
4. RT_150	Kendall's Tau B	0.278	0.500	0.389	—
	p-value	0.358	0.075	0.180	—

Appendix D

Conover's Post Hoc Comparisons, Non-Matching Stimuli, Leap Condition

Conover's Post Hoc Comparisons - Angle

		T-Stat	df	W _i	W _j	p	P _{bonf}	P _{holm}
0	50	2.337	24	11.000	24.000	0.028	0.169	0.112
	100	3.236	24	11.000	29.000	0.004	0.021	0.021
	150	2.697	24	11.000	26.000	0.013	0.076	0.063
50	100	0.899	24	24.000	29.000	0.378	1.000	1.000
	150	0.360	24	24.000	26.000	0.722	1.000	1.000
100	150	0.539	24	29.000	26.000	0.595	1.000	1.000

Note. Grouped by subject.

Conover's Post Hoc Comparisons, Non-Matching Stimuli, Mental Rotation Condition

Conover's Post Hoc Comparisons - Angle

		T-Stat	df	W _i	W _j	p	P _{bonf}	P _{holm}
0	50	0.539	24	15.000	12.000	0.595	1.000	1.000
	100	2.697	24	15.000	30.000	0.013	0.076	0.038
	150	3.236	24	15.000	33.000	0.004	0.021	0.018
50	100	3.236	24	12.000	30.000	0.004	0.021	0.018
	150	3.776	24	12.000	33.000	< .001	0.006	0.006
100	150	0.539	24	30.000	33.000	0.595	1.000	1.000

Note. Grouped by subject.

Appendix E

Correlations, Non-Matching Stimuli, Analytic Condition

Kendall's Tau Correlations

Variable		RT_0	RT_50	RT_100	RT_150
1. RT_0	Kendall's Tau B	—			
	p-value	—			
2. RT_50	Kendall's Tau B	0.389	—		
	p-value	0.180	—		
3. RT_100	Kendall's Tau B	0.222	0.722	—	
	p-value	0.477	0.006	—	
4. RT_150	Kendall's Tau B	-0.056	0.444	0.611	—
	p-value	0.919	0.119	0.025	—

Correlations, Non-Matching Stimuli, Leap Condition

Kendall's Tau Correlations

Variable		RT_0	RT_50	RT_100	RT_150
1. RT_0	Kendall's Tau B	—			
	p-value	—			
2. RT_50	Kendall's Tau B	0.667	—		
	p-value	0.013	—		
3. RT_100	Kendall's Tau B	0.444	0.667	—	
	p-value	0.119	0.013	—	
4. RT_150	Kendall's Tau B	0.333	0.333	0.556	—
	p-value	0.260	0.260	0.045	—

Correlations, Non-Matching Stimuli, Mental Rotation Condition

Kendall's Tau Correlations ▼

Variable		RT_0	RT_50	RT_100	RT_150
1. RT_0	Kendall's Tau B	—			
	p-value	—			
2. RT_50	Kendall's Tau B	0.833	—		
	p-value	< .001	—		
3. RT_100	Kendall's Tau B	0.722	0.778	—	
	p-value	0.006	0.002	—	
4. RT_150	Kendall's Tau B	0.556	0.611	0.611	—
	p-value	0.045	0.025	0.025	—